



HIGHER-ORDER DERIVATIVES

Basic Calculus

Higher-order Derivatives

- Note that one consequence of having the derivative function is that you can further compute the derivative of the derivative. As it turns out, such higher-order derivatives have important applications. Suppose that you start with a function f and compute its first derivative f' . You can still compute its second derivative f'' , its third derivative f''' , and so on. Table 11.1 gives other notations needed to familiarize when computing higher-order derivatives.

Table 11.1. Notations to be used when computing higher-order derivatives

	Prime Notation	Function Notation	Other Notations
First derivative			
Second derivative			
Third derivative			
Fourth derivative			

Example 1:

- Find the first five derivatives of $f(x) = 5x^4 - 3x^3 + 1$

Solution:

$$\bullet f'(x) = \frac{dy}{dx} = 20x^3 - 9x^2$$

$$\bullet f''(x) = \frac{d^2y}{dx^2} = 60x^2 - 18x$$

$$\bullet f'''(x) = \frac{d^3y}{dx^3} = 120x - 18$$

$$\bullet f^{(4)}(x) = \frac{d^4y}{dx^4} = 120$$

$$\bullet f^{(5)}(x) = \frac{d^5y}{dx^5} = 0$$

Example 2:

- Find all the derivatives of the function $f(x) = 8x^4 + 5x^3 - x^2 + 7$

Solution:

- $f'(x) = 32x^3 + 15x^2 - 2x$
- $f''(x) = 96x^2 + 30x - 2$
- $f'''(x) = 192x + 30$
- $f^{(4)}(x) = 192$
- $f^{(5)}(x) = 0$

Big Idea:

- Higher-order derivatives involve successive differentiation. A higher-order derivative translates into a series of change functions derived from a single function. In a similar way, you often encounter a successive changes of things around you, which lead to improvements. Change is the only thing constant in the world. You have to find ways to keep changing for the better.

Example 3:

- Find the third derivative of $y = \frac{2x}{3x-5}$

Solution:

• Let $u = 2x$ and $v = 3x - 5$ so that $du = 2$ and $dv = 3$. Using quotient rule, you have

$$\bullet y' = \frac{(3x-5)(2)-(2x)(3)}{(3x-5)^2} = \frac{6x-10-6x}{(3x-5)^2} = \frac{-10}{(3x-5)^2}$$

Solution:

- To find the second derivative, let $u = -10$ and $v = (3x - 5)^2$. This means that $du = 0$ and $dv = 2(3x - 5)(3) = 6(3x - 5)$.

$$\bullet y'' = \frac{(3x-5)^2(0) - (-10)[6(3x-5)]}{(3x-5)^4} = \frac{60(3x-5)}{(3x-5)^4} = \frac{60}{(3x-5)^3}$$

Solution:

- To find the third derivative, let $u = 60$ and $v = (3x - 5)^3$ giving $du = 0$ and $dv = 3(3x - 5)^2(3) = 9(3x - 5)^2$. Then,

$$\bullet y''' = \frac{(3x-5)^2(0) - (60[9(3x-5)^2])}{(3x-5)^6} = \frac{-540(3x-5)^2}{(3x-5)^6} =$$

$$-\frac{540}{(3x-5)^4}$$